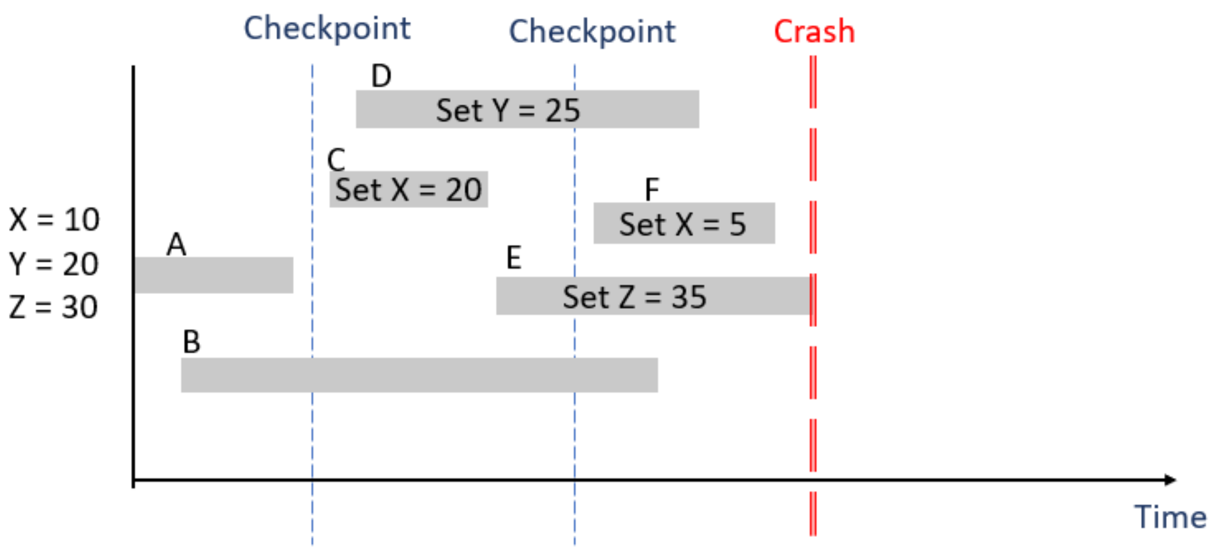
Assume that a DBMS has several transactions running with the timeline depicted in the figure being executed. You can also assume that each transaction at the end will modify the variables as shown in the figure. The DBMS crashes as shown.



Assume that the DBMS uses immediate update protocol:

a. Which transactions, if any, need to be redone?
   B,C,D,F
   b. Which transactions, if any, need to be undone?
   E
   c. Which transactions, if any, are not affected by the crash?
   A
   d. What will the final value of x,y,and z be in the database after everything is restored?
   X=5,y=25,Z=30

Now assume that the DBMS uses the deferred update protocol:

a. Which transactions, if any, need to be redone?
   B,C,D,F
   b. Which transactions, if any, need to be undone?
   None
   c. Which transactions, if any, are not affected by the crash?
   A and E
   d. What will the final value of x,y,and z be in the database after everything is restored?
   X=5,Y=25,Z=30

2. Using your Oracle database (the instance that you installed for Homework 01), write SQL commands to create the following tables and then answer each of the questions:

Student(StudentId, StudentFName, StudentLName, Major, Credits)
Faculty(FacultyId, FacultyName, Department, Rank)
Class(ClassId, FacultyId, Timeslot, RoomNbr)
Enroll(StudentId, ClassId, Grade)

```
CREATE TABLE Student (
   StudentId INT PRIMARY KEY,
   StudentFName VARCHAR2(50),
   StudentLName VARCHAR2(50),
   Major VARCHAR2(50),
   Credits INT
);

CREATE TABLE Faculty (
   FacultyId INT PRIMARY KEY,
   FacultyName VARCHAR2(100),
   Department VARCHAR2(50),
   Rank VARCHAR2(50)
);

CREATE TABLE Class (
   ClassId INT PRIMARY KEY,
   FacultyId INT,
   Timeslot VARCHAR2(50),
   RoomNbr VARCHAR2(20)
);

CREATE TABLE Enroll (
   StudentId INT,
```

```
  ClassId INT,
  Grade VARCHAR2(2),
  PRIMARY KEY (StudentId, ClassId)
);
```

a. Create a user Tom201 and give them permissions to read and update the student class enrollments, however they should not have access to the student's grade. How can you do that? Hint: Create a view

```
CREATE USER Tom201 IDENTIFIED BY pass123;
GRANT CREATE SESSION TO Tom201;
CREATE VIEW Student_Enrollments AS
SELECT StudentId, ClassId
FROM Enroll;
GRANT SELECT, UPDATE ON Student_Enrollments TO Tom201;
```

b. Allow everyone who works in the Dean's Office to read student data.
Hint: Create a role named DeansOffice that has full read access to the Student table – Grant this to every employee in the Dean's office. First you need to create a user for one of the employees in the Dean's office.

```
CREATE ROLE DeansOffice;
GRANT SELECT ON Student TO DeansOffice;
CREATE USER DeanStaff1 IDENTIFIED BY pass123;
GRANT CREATE SESSION TO DeanStaff1;
GRANT DeansOffice TO DeanStaff1;
```

c. Revoke the privileges of part a.

```
REVOKE SELECT, UPDATE ON Student_Enrollments FROM Tom201;
```